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RESEARCH ARTICLE

An Efficient Algorithm for Designing Long Aperiodic Binary Sequences With Low Auto-Correlation Sidelobes

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ABSTRACT Sequences with low auto-correlation sidelobes are important in diverse fields, including communications, radar, sonar, and medical imaging. The task of finding binary sequences with low auto-correlation is known as a difficult optimization problem. Recently, many computational methods have been proposed for designing long binary sequences with good auto-correlation properties. In this paper, we present a new algorithm to generate long aperiodic binary sequences with low peak sidelobe levels. Our algorithm combines the Legendre sequence to seed an initial binary sequence, and an efficient stochastic search method with a dynamic fitness function mechanism to generate a final binary sequence. Numerical examples of the obtained binary sequences with a length N up to $2^{20} - 1$ and a peak sidelobe level lower than $0.8\sqrt{N}$ are provided, to demonstrate the efficiency of the proposed stochastic algorithm, that can outperform existing computational methods for designing long aperiodic binary sequences.

INDEX TERMS Auto-correlation functions, binary sequences, long aperiodic sequences, peak sidelobe level (PSL).

I. INTRODUCTION

Finding sequences with low auto-correlation sidelobes is a very challenging optimization problem of the digital sequence design. Such sequences are used widely in various practical fields, such as radar applications, telecommunications, cryptography, active sensing and communications systems [1], [2], [3], [4], [5], [6], [7], [8], [9]. Nowadays, they are especially useful in autonomous driving vehicles [6], where multiple automotive radar sensors are needed. Recently, several algorithms were applied to solve this challenging optimization problem, for example, the quantum approximate optimization algorithm [10], the EMAS meta-heuristic with various local optimization techniques [11], etc. In [12], the authors present a complex networks analysis of the energy landscape of the low auto-correlation binary sequences problem. The problem has a connection to Littlewood polynomials – polynomials with plus and minus one

coefficients – and, recently, an excellent paper has been published, ‘Flat Littlewood polynomials exist’ [13].

Autocorrelation functions have two different definitions – periodic and aperiodic autocorrelations [14]. Traditionally, the periodic autocorrelation [15] of a binary sequence has received more attention in literature for sequence families of CDMA communication systems [16]. However, the aperiodic autocorrelation is considered better, to characterize a binary sequence for more realistic communication systems [16].

Here, we describe the definitions of a binary sequence, a low auto-correlation binary sequence problem, and two metrics. A binary sequence $S = (s_0, s_1, \dots, s_{N-1})$ of length N has all entries either -1 or $+1$. The aperiodic auto-correlation function of S at shift k is defined as:

$$\begin{aligned} r(k) &= \sum_{i=0}^{N-k-1} s_i s_{i+k} \\ &= r(-k), \quad \text{for } k = 0, 1, \dots, N-1. \end{aligned} \quad (1)$$

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The *Peak Sidelobe Level* (PSL) metric of S is defined as follows:

$$\text{PSL} = \max_{1 \leq k \leq N} |r(k)|. \quad (2)$$

The auto-correlation function of S at shift 0, $r(0)$, is called the *main-lobe* level, and it is not included in (2). The rest, $r(k)$, $k = 1, 2, \dots, N - 1$, are called *side-lobe* levels. The aperiodic auto-correlation function in (1) is an even function, since $r(k) = r(-k)$, and therefore, it is enough to consider it for the interval $k = 0, 1, \dots, (N - 1)$ only. The PSL value in decibels, denoted as NPSL, is represented as:

$$\text{NPSL [dB]} = 20 \log_{10} \left(\frac{\text{PSL}}{N} \right). \quad (3)$$

The *Integrated Sidelobe Level* (ISL) is another measure of the smallness of the aperiodic auto-correlation functions:

$$\text{ISL} = \sum_{k=1}^{N-1} |r(k)|^2. \quad (4)$$

Note that ISL is defined as the sum of the squares of all the off-peak auto-correlations.

The *Low Auto-correlation Binary Sequence* (LABS) problem involves assigning values to the s_i that minimize ISL or maximize the *merit factor* F [17], which is calculated as follows:

$$F = \frac{N^2}{2 \cdot \sum_{k=1}^{N-1} |r(k)|^2} = \frac{N^2}{2 \cdot \text{ISL}}. \quad (5)$$

The merit factor is a measure of the quality of the sequences in terms of engineering applications [18]. A sequence with the optimal PSL usually has a merit factor which is much lower than the optimal merit factor, and vice versa [19]. Therefore, it can be useful that an optimization algorithm applies both measures simultaneously in the early stages of the optimization process when searching for a good binary sequence, but, in later stages, the algorithm needs to focus on applying a single measure only. In this paper, we are focused mainly on generating binary sequences with low PSL values, but we will also report the merit factor values of the obtained binary sequences.

The main goal of this work is to design long binary sequences with low PSL values through a computational approach using a stochastic algorithm. We will use Legendre sequences to initialize (seed) our algorithm, and, for clarity, we give their brief definitions. Let p be a prime number. A *Legendre sequence* is a binary sequence with respect to the prime p and it is defined by:

$$s_i = \left(\frac{i}{p} \right), \quad \text{for } 0 \leq i < p, \quad (6)$$

where $\left(\frac{i}{p} \right)$ is the Legendre symbol [20], [21]. The Legendre symbol takes the value 1 if i is a quadratic residue modulo p and the value -1 if it is not. We use the convention that $\left(\frac{i}{p} \right) = 1$ if $i = 0$.

Through extensive experimental work, we investigate a growing rate of PSL values in regard to the lengths of sequences. In this work, we use an improved version of a stochastic algorithm for finding aperiodic binary sequences. The contributions of this paper are:

- A new stochastic algorithm is proposed for finding aperiodic binary sequences with low PSL values. It combines the Legendre sequence, and an efficient stochastic search method with a dynamic fitness function mechanism.
- The new best-known PSL values are obtained by the algorithm for binary sequences of length up to $2^{20} - 1 = 1\,048\,575$.
- All of the obtained binary sequences have $\text{PSL} < 0.8\sqrt{N}$.

The rest of this paper is organized as follows. The literature review is given in Section II. The proposed stochastic algorithm is presented in Section III. In The experimental results are reported in Section IV. A brief discussion is given in Section V. Finally, the paper ends with a conclusion and future work in Section VI.

II. LITERATURE REVIEW

Since the 1950s, construction methods have been considered to design binary sequences with low auto-correlation sidelobes [6]. Barker codes have excellent aperiodic auto-correlation properties, but the size of the longest known Barker code is 13 [20], which is usually too short for practical applications. Legendre sequences [22] have low aperiodic auto-correlation sidelobes, but their lengths need to be prime numbers. In [23] the author proposed a construction of aperiodic sequences with PSL values not greater than $\sqrt{2N \ln(2N)}$, and their PSL values increase in the order of $\sqrt{N \ln(\ln N)}$. In works [2], [5], an algorithm for construction of binary sequences whose aperiodic PSL grows like $O(\sqrt{N})$ is proposed, and several numerical examples are presented with length $N \sim 2^{12}$. Several other construction methods exist. None of the construction methods guarantees that the generated binary sequences will possess the lowest possible (optimal) PSL value [24]. Using an exhaustive search method is the only way to find a binary sequence with the optimal PSL value, but the exhaustive search method is only applicable for smaller sequences.

Considerable computational effort has been put into finding binary sequences with small PSL values [23], [27], [28]. Currently, binary sequences with the best-known PSL values for $86 \leq N \leq 105$ are published in [27], while results for $106 \leq N \leq 300$ are reported in [19], [24], and [29], for selected lengths for $301 < N \lesssim 4000$ in [6], [19], [29], [30], [31], [32], [33], [34], and [35], for N up to 8191 in [2] and [5], and for even longer N (up to 10^6) in [4], [6], [9], [19], and [25]. It can be noticed that many of these researches deal with binary sequences with moderately short lengths, i.e., up to 5000, and the rest of them offer only a small number of results for (very) long binary sequences. We think that the reason for this lies in the high computational complexity.

TABLE 1. The comparison with previous works.

Property	Literature				
	[25]	[9]	[19]	[26]	this work
Year published	2021	2024	2021	2024	
Problem representation (two one-dimen. arrays)	✓	–	✓	✓	✓
One bit flipping mechanism	✓	–	✓	✓	✓
Fitness function (parameter a in Eq. (8)*)	$a = 2$	$a = 2^p - 2, p \in \{1, \dots, 10\}$	$a \in \{0, \dots, 6\}$	$a_1 = 2, a_2 \in \{8, 9, 10\}$	$a \in \{0, \dots, 68\}$
Initial binary sequence	–	–	–	–	Legendre seq.
Programming language	C++	NA	C/C++	C++	C/C++
Parallel implementation	CPU, CUDA	NA	CPU	CPU + CUDA	CPU
Length of longest sequence	1 048 575	131 071	1 048 575	1 048 575	1 048 575

* – note that $a + 2$ is in Eq. (8).

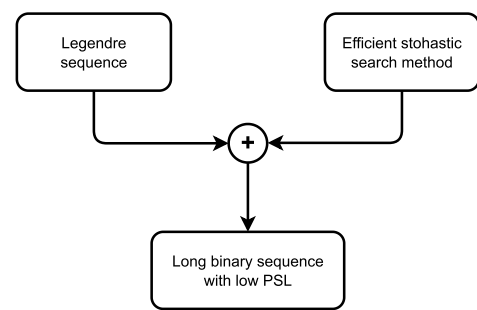
Usually, authors report only the best known PSL values in publications, so we estimate that the current best values are also (or are very close to) the optimal PSL values for binary sequences of a length approximately up to 150. For longer sequences, however, no optimal PSL values are known. Currently, the best known PSL value of 889 for binary sequence of length $2^{20} - 1$ is reported in [26], and the previous reported PSL of 1024 [25] has been improved by 135. In this work, we generate a binary sequence with a PSL equal to 770.

When solving the LABS problem using either the PSL or F metric, an efficient implementation of the aperiodic auto-correlation [Eq. (1)] is needed for a single bit flipping in a binary sequence. In the literature, researchers have presented such calculations [4], [29], [35], [36], [37], [38].

In this work, we use an improved version of a stochastic algorithm for finding aperiodic binary sequences. The new algorithm is based on the works of Dimitrov et al. [29] and our work [19]. Table 1 gives an overview to compare this work with previous related works. One can see that the efficient one bit flipping mechanism is used by all works except in [9], where the fast Fourier transform (FFT) is used with the computational complexity of $O(N \log_2 N)$ (note that this work used an efficient line search procedure too). All works differ a lot on how to use parameter a , [25] uses one value of $a + 2 = 2 + 2 = 4$, [26] applies two phases, the first one with $a_1 = 2$ and the second one where a_2 is selected from a set $\{8, 9, 10\}$ based on the sequence's length, the algorithm proposed in [9] uses $a = 2^p - 2$ where it starts with $p = 2$ and increases p gradually until $p = 2^{10}$, while [19] and this work use the parameter a from $\{0, \dots, 6\}$ and $\{0, \dots, 68\}$, respectively. Only this work utilizes the Legendre sequence during a sequence initialization, while several works used the C/C++ programming language without any additional library. Parallel implementation is only applied in [25] and [26].

III. THE PROPOSED ALGORITHM

In this section, we present a new heuristic algorithm for generating binary sequences with good auto-correlation properties computationally.

**FIGURE 1.** Illustrated the main concept of our algorithm.

Our heuristic algorithm uses an implementation of one bit flipping as proposed in our previous work [19], which is based on the mechanism proposed by M. Dimitrov et al. [29], and, for clarity, we present a procedure of this mechanism in Alg. 1. This mechanism uses two one-dimensional arrays, S and Ω_S , to store a binary sequence, and its sidelobe levels, respectively.

Fig. 1 illustrates the main concept of our algorithm. The pseudo-code of the proposed algorithm is presented in Alg. 2. The main characteristics of the algorithm are as follows. A binary sequence is initialized with a 1/4-rotated Legendre sequence [using (6)], which is then truncated, if needed, to ensure that the length of the binary sequence is equal to N . Note that none of the algorithms published in [19] and [29] use such initialization.

During the main loop, our algorithm is generating a binary sequence with a low PSL value throughout flipping bits, and inside the inner loop (i.e., the *for loop* which starts in Line 10), a single bit in the binary sequence is flipped one after the other, and checked if the Cost value in Line 18 is improved. If it is improved, a new cycle of the inner loop is started, otherwise an undo flip operation is performed.

Our stochastic algorithm starts flipping operations on the initial sequence at place s_r , where $r = 3N/4$ (Line 7 in Alg. 2), and continues flipping adjacent bits in the sequence one after another. In a later stage of the optimization process,

Algorithm 1 One bit flip of a binary sequence [29]

```

**  $f$  ...index of flipping bit;  $S$  ...binary sequence **
**  $\Omega_S$  ...sidelobe levels;  $N$  ...sequence length **
1: procedure Flip( $f$ ,  $S$ ,  $\Omega_S$ ,  $N$ )
2:  $\delta_{min} \leftarrow \min(N - f - 1, f)$ 
3:  $\delta_{max} \leftarrow \max(N - f, f)$ 
4: if  $f \leq \frac{N-1}{2}$  then
5:   for  $q \in [0, \delta_{max} - \delta_{min} - 1]$  do
6:      $\Omega_S[\delta_{min} + q] -= 2S[f]S[N - q - 1]$ 
7:   end for
8: else
9:   for  $q \in [0, \delta_{max} - \delta_{min}]$  do
10:     $\Omega_S[\delta_{min} + q] -= 2S[f]S[q]$ 
11:   end for
12: end if
13: if  $f \leq \frac{N-1}{2}$  then
14:   for  $q \in [0, N - \delta_{max}]$  do
15:     $\Omega_S[\delta_{max} + q - 1] -= 2S[f](S[2f - q] + S[q])$ 
16:   end for
17: else
18:   for  $q \in [0, N - \delta_{max} - 1]$  do
19:     $\Omega_S[\delta_{max} + q] -=$ 
20:     $2S[f](S[\delta_{max} - \delta_{min} + q] + S[N - q - 1])$ 
21:   end for
22:  $S[f] = -S[f]$  ** flip bit at index  $f$  **
23: end procedure

```

the algorithm stores the last flipped bit in Lines 16 and 20, and uses it in the next iteration of the inner loop. Both previous algorithms published in [19] and [29] do not use such a mechanism of storing the last flipped bit.

A fitness function $F(\Omega_S)$ is incorporated in both of the previous mentioned algorithms, and it is also used in our algorithm, but the fitness function for calculating a Cost value differs a lot in each algorithm. In [29], $F_a(S) = \sum_{k=1}^{N-1} |r(k)|^4$ is used, while (7) is applied in [19].

$$F_a(S) = \begin{cases} \sum_{k=1}^{N-1} |r(k)|^{a+3}, & \text{for } a \in \{0, 2, 4\}, \\ \sum_{k=1}^{N-1} (|r(k)|^{a+3} + |r(k)|^{a+2}), & \text{for } a \in \{1, 3, 5\}. \end{cases} \quad (7)$$

In this work, we will use a fitness function shown in (8). The value a (Line 38) is being changed during each small perturbation of the current binary sequence. The proposed dynamic fitness function mechanism allows our algorithm to generate binary sequences more efficiently. Using parameter a enables a more intensive effect on the perturbations of the binary sequence during the optimization process, and, thus, a better binary sequence can be obtained.

$$F_a(S) = \sum_{k=1}^{N-1} |r(k)|^{a+2}, \quad \text{for } a \in \{0, 1, \dots, 68\}. \quad (8)$$

Algorithm 2 Improved algorithm for generating aperiodic binary sequences with low PSL.

```

Require:  $N$  ** length of a binary sequence **
Ensure:  $S$  ** optimized binary sequence with length equal to  $N$  **
Ensure: BestPSL ** PSL value of the optimized binary sequence  $S$  **
** Sequence initialization **
1: Find lowest prime  $p$  that is equal or greater than  $N$ 
2: Generate  $S$  using  $p$  and Eq. (6) ** initialize Legendre sequence **
3: Truncate  $S$  to length  $N$  if  $p > N$ 
4: Rotate  $S$  for a fraction of  $w = 1/4$  ** initial sequence **
5: BestCost, BestPSL  $\leftarrow F(\Omega_S)$  ** calculate fitness function **
6: isGImpr, isLImpr  $\leftarrow$  true, false
7:  $r \leftarrow (1 - w)N$  ** first bit to be flipped **
** main optimization loop **
8: while true do
9:   if isGImpr then
10:    ** inner loop **
11:    for ( $i \leftarrow 0$ ;  $i < N$ ;  $i++$ ) do
12:       $f \leftarrow (r + i) \% N$ 
13:      Flip( $f$ ,  $S$ ,  $\Omega_S$ ,  $N$ ) ** flip single bit **
14:      Cost, PSL  $\leftarrow F(\Omega_S)$  ** calculate fitness function **
15:      if PSL < BestPSL then
16:        BestPSL  $\leftarrow$  PSL ** store the best PSL value **
17:         $r \leftarrow f$ 
18:      end if
19:      if Cost < BestCost then
20:        BestCost, isLImpr  $\leftarrow$  Cost, true
21:         $r \leftarrow f$ 
22:        break
23:      else
24:        Flip( $f$ ,  $S$ ,  $\Omega_S$ ,  $N$ ) ** undo flip **
25:      end if
26:    end for
27:    if isLImpr then
28:      isGImpr, isLImpr  $\leftarrow$  true, false
29:      continue
30:    else
31:      isGImpr  $\leftarrow$  false
32:    end if
33:    ** Small perturbation: randomly flip  $k \in [1, 2]$  bits **
34:     $k \leftarrow 1 + \text{Rnd}(2)$ 
35:    for ( $j \leftarrow 0$ ;  $j < k$ ;  $j++$ ) do
36:       $f \leftarrow \text{Rnd}(N)$ 
37:      Flip( $f$ ,  $S$ ,  $\Omega_S$ ,  $N$ ) ** flip single bit **
38:    end for
39:     $a \leftarrow \text{Rnd}(69)$  **  $a \in \{0, 1, \dots, 68\}$  **
40:    BestCost, PSL  $\leftarrow F(\Omega_S)$  ** calculate fitness function **
41:    if PSL < BestPSL then
42:      BestPSL  $\leftarrow$  PSL ** store the best PSL value **
43:    end if
44:    isGImpr, isLImpr  $\leftarrow$  true, false
45:  end while

```

Note that $F_a(S)$ is used for calculating a Cost value, while (2) is used for calculating a PSL value, and both calculations are combined in $F(\Omega_S)$.

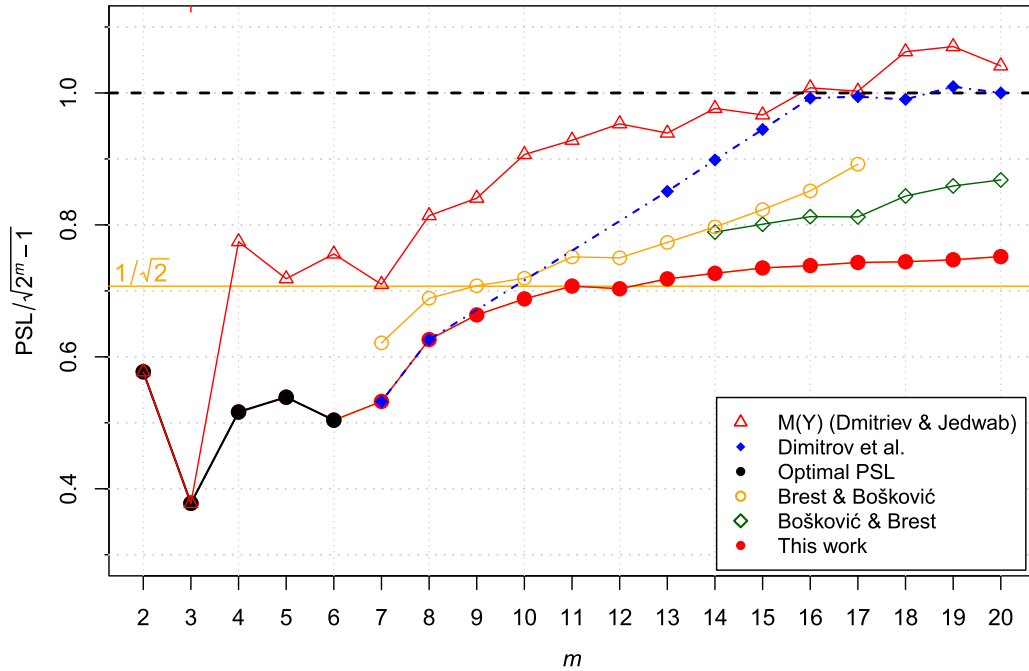


FIGURE 2. Growth rate of PSL with $\sqrt{N}$, where $N = 2^m - 1$. PSL/ $\sqrt{N}$ versus m is presented, and lower values are better. The results for M(Y) (and W(Y) for $m \geq 18$) are taken from [39], Dimitrov et al. [24], [25], [29], Brest & Bošković [19], Bošković & Brest [26].

TABLE 2. PSL values of initial and optimized binary sequences with lengths up to $2^{20} - 1$.

m	$N = 2^m - 1$	Initial	Optimized		
		PSL	PSL (NPSL)	F	
11	2047	49	32 (-36.12 dB)	4.83	
12	4095	57	45 (-39.18 dB)	5.10	
13	8191	88	65 (-42.01 dB)	5.11	
14	16383	125	93 (-44.92 dB)	5.04	
15	32767	183	133 (-47.83 dB)	5.00	
16	65536	303	189 (-50.80 dB)	5.05	
17	131071	392	269 (-53.74 dB)	5.00	
18	262143	625	381 (-56.75 dB)	5.03	
19	524287	808	541 (-59.73 dB)	4.99	
20	1048575	1172	770 (-62.68 dB)	5.00	

TABLE 3. A collection of the NPSL, PSL, and F values of binary sequences with length $N = 10^6$.

Ref.	NPSL	PSL	F
[19]	-58.98 dB	1125	NA
[4]	-56.1 dB	$\approx 1566^*$	NA
[6]	-56.32 dB	$\approx 1527^*$	4.94
this work	-62.35 dB	763	4.91

* – a PSL value is calculated using (3).

IV. EXPERIMENTAL RESULTS

The PSL values of the initial and optimized binary sequences with lengths $N = 2^m - 1$, $m = 11, 12, \dots, 20$ are shown in

Table 2. An initial binary sequence is constructed from the 1/4-rotated Legendre sequence. The initial sequence is used as a seed by the proposed algorithm, yielding an optimized final binary sequence. The NPSL values in decibels and merit factors of the optimized binary sequences are shown in the last two columns. Interestingly, all the merit factor values are approximately equal to 5.

Fig. 2 shows the growth rate of the obtained PSL values with $\sqrt{N}$, where $N = 2^m - 1$ for $m = 2, 3, \dots, 20$. Values PSL/ $\sqrt{N}$ versus m are depicted in this Figure. Dimitriev & Jedwab [39] studied the growing rate of PSL values of binary sequences, and values M(Y) (and W(Y) for $m \geq 18$) are taken from [39]. The next results are combined from three papers: Dimitrov et al. [24], [29], and Dimitrov [25], and it can be seen that these PSL results for $m = 16, 17, \dots, 20$ are close to $\sqrt{N}$ (i.e., PSL/ $\sqrt{N}$ is close to the dash line with 1.0). For shorter sequences there are no differences between the algorithms, except the M(Y) values, which are higher than other values – note that the M(Y) values are obtained by using a different approach. For larger sequences, for $m > 10$, a difference between the algorithms is noticeable. Our previous works (Brest and Bošković [19] and Bošković and Brest [26]) reported results of $0.8 < \text{PSL}/\sqrt{N} < 0.9$.

The obtained results PSL/ $\sqrt{N}$ of the proposed algorithm (PSL values are shown in Table 2) are lower than 0.8, and they are lower than, or close to $1/\sqrt{2}$ for $m = 8, 9, \dots, 12$, while, for $m > 13$ they have a slowly increasing trend, as can be seen in Fig. 2. The latter also indicates that, if the size of the binary sequence is increasing, the optimization problem

of finding a binary sequence with a low PSL value becomes harder.

Table 3 presents a collection of the NPSL values, reported by the state-of-the-art algorithms, for binary sequences with $N = 10^6$. Note that the collected results must be taken with a special care, since they were performed on different computer machines and computational time conditions. There are two interesting things that can be observed from the results in this Table. First, only the last result of the PSL value is lower than $\sqrt{N} = 1000$, and second, both reported merit factors are ≈ 5 .

To compare with the result in [9], we chose $N = 2^{17} - 1 = 131071$. The PSL value and merit factor (F) of the sequence in this work are 269 and 5.00, respectively, whereas they are 284 and 4.59 in [9].

Our algorithm was implemented using the C/C++ programming language without additional libraries. The computational experiments were performed on a Linux Mint with i7-4770 CPU 3.40 GHz and on a grid for sequences with $m \leq 17$. The binary sequences obtained by our algorithm are collected and available on <https://github.com/J-Brest/LABS>.

V. DISCUSSION

Legendre sequences are only available for prime numbers, but, in this work, we used the Legendre sequences that are shortened to become an equal size to our desired length. For example, the primes in Table 2 are only 8191, 131 071, 524 287, while all other lengths and 10^6 are not primes. Therefore, our algorithm is applicable for any length.

The 1/4-rotated Legendre sequences have $F = 6$ [18], [20]. The obtained binary sequences with low PSL have $F \approx 5$, and a binary sequence with $F = 4.94$ is also reported in [6]. We have not yet found any research in the literature that addresses what merit factors have binary sequences that are optimal or near-optimal in the sense of the PSL metric.

How the proposed generated binary sequences could be integrated into some system (e.g., in cryptography) represents an additional challenge, and further work can continue in this direction. We highlight just a few articles where the reader can find guidelines for using binary sequences with low autocorrelation functions: [6], [8], [40], [41] and the references therein.

VI. CONCLUSION

Finding low auto-correlation binary sequences with good peak sidelobe level (PSL) values is a difficult optimization problem. In this study, we proposed a new heuristic algorithm for designing aperiodic binary sequences of length up to $2^{20} - 1 = 1\,048\,575$ with low PSL values. Our algorithm combines the Legendre sequence to seed an initial binary sequence, and an efficient stochastic search method with a dynamic fitness function mechanism to generate a final binary sequence. We can see that the proposed heuristic algorithm generated binary sequences with $\text{PSL} \lesssim (1/\sqrt{2})\sqrt{N}$ for $m = 8, 9, \dots, 12$ where $N = 2^m - 1$.

The most important result of this work is as follows. The new best-known PSL values are obtained by the algorithm for binary sequences of length up to $2^{20} - 1$. All the PSL values of the obtained binary sequences are lower than $0.8\sqrt{N}$. Furthermore, the obtained PSL values have an increasing trend, but it can be indicated that binary sequences of length $m = 21, 22, \dots$ that have $\text{PSL} < \sqrt{N}$ probably exist.

Interestingly, the merit factors ≈ 5 have been observed by all the obtained binary sequences, and this may be a direction for future research. Our main computational result demonstrates the existence of binary sequences with $\text{PSL} < 0.8\sqrt{N}$ for N up to $2^{20} - 1$.

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